

Professional Summary

Motivated Mechanical Engineering graduate with a strong foundation in core mechanical engineering principles, currently pursuing a Bachelor’s degree in Mechatronics Engineering at Selcuk University. Experienced in hands-on engineering projects involving 3D printing, robotics, and unmanned aerial vehicle (UAV) design and development. Proficient in CAD design using SolidWorks and Fusion 360, as well as embedded systems based on Arduino platforms. Completed internships in production and financial management departments within a food machinery manufacturing company. Highly interested in computer-aided design and intelligent systems, with a strong commitment to continuous technical development and effective teamwork. Fluent in Arabic and Turkish.

Education

Selcuk University	Konya, Türkiye
Bachelor of Science in Mechatronics Engineering	Sep 2024 – Present
Selcuk University	Konya, Türkiye
Bachelor of Science in Mechanical Engineering	Oct 2019 – Jul 2023
Salfit Palestinian Secondary School	Salfit, Palestine
High School Certificate (Tawjihi), Scientific Stream	Jun 2017 – Jul 2018

Projects

3D Printing Systems Study	2022
- Conducted a comprehensive study on 3D printing technologies, including hardware components, slicing software, and printing materials - Analyzed mechanical performance, print quality parameters, and structural limitations - Modeled system architecture and evaluated potential industrial applications	
Hybrid Ground–Aerial Vehicle	2022
- Led a research-based design project for a hybrid vehicle capable of ground mobility and short-range flight - Performed aerodynamic calculations and structural feasibility analysis - Evaluated propulsion system configurations and future mobility applications	
Industrial Robotics Research Project	2023
- Conducted an in-depth study of industrial robotic systems used in manufacturing environments - Analyzed robot classifications, control systems, kinematic structures, and sensor technologies - Investigated real-world industrial applications and future trends in smart factories and automation	
RRP Robotic Arm Project (Revolute–Prismatic–Prismatic)	2024
- Assembled an RRP-type robotic manipulator using 3D-printed components - Integrated servo motor actuation system - Developed complete kinematic modeling including joint variable calculations and transformation matrices - Implemented forward and inverse kinematics analysis for motion planning - Prepared a detailed technical report covering mechanical constraints, testing procedures, and system validation	

Autonomous Obstacle Avoidance Smart Car	2024
- Designed and fabricated a 3D-printed autonomous mobile robot - Integrated Arduino controller and ultrasonic sensors for obstacle detection - Implemented real-time decision-making algorithm enabling automatic path correction - Developed specifically for robotics and obstacle-challenge competitions	
Smart Hospital Assistance Robot	2025
- Assembled and programmed a compact hospital assistance robot - Integrated sensors for heart rate monitoring, blood quality indicators, and air quality measurement - Implemented wireless communication for real-time data transmission - Developed system architecture to support hospital monitoring environments	
Medical Delivery Drone	2025
- Designed and developed a medical UAV using SolidWorks and Fusion 360 - Engineered drone structure optimized for payload delivery to remote areas - Integrated autonomous navigation system and servo-controlled payload release mechanism - Presented at TEKNO-SEL 2025 Technology Exhibition as an emergency medical solution	
Disaster Relief UAV System (Ongoing)	2025
- Designed and built a UAV platform for post-disaster operations such as earthquakes and floods - Developed mechanical structure using SolidWorks and modular assembly approach - Planned integration of vision-based human detection system using onboard cameras - Proposed AI-based image processing for victim identification - Implemented emergency location and image transmission to rescue teams	
Single-Axis Reaction Wheel Attitude Control System	2026
- Fabricated and assembled a single-axis reaction wheel system using 3D printing technology - Implemented angular momentum–based attitude stabilization concept - Integrated NEMA-17 motor, MPU6050 IMU, and Arduino Nano controller - Developed PD and PID control algorithms for attitude stabilization and detumbling - Experimentally validated disturbance rejection performance and reaction-wheel saturation behavior	
4-Axis Robotic Arm with Modular Gripper	2026
- Fabricated and assembled a 3D-printed four-axis robotic arm - Integrated high-torque servo motors and a modular end-effector system - Implemented internal cable routing to improve mechanical reliability and system organization - Developed motion control system using Arduino and servo driver modules - Enabled precise multi-axis motion for robotics experimentation and educational automation	

Work Experience	
Mechanical Design Engineer	Konya, Türkiye
SRT Gear Manufacturing Company	Feb 2024
- Designed gear systems and automotive spare parts - Prepared technical drawings and 3D models - Supported design optimization and manufacturing documentation	

Internships

Production Engineering Intern

Efor Machinery

Konya, Türkiye

Jun 2022 – Aug 2022

- Participated in food processing machinery manufacturing operations
- Observed machining, assembly, and production workflows
- Gained hands-on exposure to industrial materials and electrical systems

Financial Management Intern

Efor Machinery

Konya, Türkiye

Aug 2022 – Sep 2022

- Assisted in basic financial and cost-management operations
- Gained understanding of industrial budgeting and reporting

Mechanical Engineering Orientation Training Program

Selcuk University

Konya, Türkiye

Jun 2021 – Aug 2021

- Completed industrial orientation program for mechanical engineering students
- Learned professional roles and responsibilities of mechanical engineers

Leadership & Activities

TEKNO-SEL Technology Exhibition

Project Presenter

Konya, Türkiye

May 2025

- Presented a medical delivery drone project as an innovative emergency healthcare solution
- Explained system architecture, payload release mechanism, and autonomous navigation concept
- Communicated with technical evaluators and industry representatives

University Engineering Projects Team

Project Lead

Konya, Türkiye

2023 – Present

- Collaborated on multidisciplinary engineering projects involving robotics and UAV systems
- Contributed to system integration, testing, and technical documentation
- Participated in project demonstrations and academic presentations

Self-Directed Engineering Projects

Independent Developer

Konya, Türkiye

2021 – Present

- Conducted continuous hands-on engineering projects in robotics, drones, and embedded systems
- Managed full project lifecycle including assembly, testing, troubleshooting, and documentation
- Maintained technical development through independent research and experimentation

Skills & Interests

- **Technical:** SolidWorks (Advanced), Fusion 360 (Advanced), AutoCAD (Intermediate), Mechanical Design, Drone Design, MATLAB (Intermediate), Proteus, Microsoft Office (Word, Excel, PowerPoint)
- **Language:** Arabic (Native), Turkish (Advanced), English (Intermediate)
- **Laboratory:** 3D Printing (PLA, PETG), Mechanical Assembly, Prototyping, Sensor Integration (IMU, GPS, Ultrasonic), Embedded Systems Testing and Calibration
- **Interests:** UAV Systems, Robotics, Autonomous Systems, Additive Manufacturing, Computer Vision

Certifications

Microsoft Office (Word, Excel, PowerPoint)	2023
AutoCAD	2023
Basic Technical Drawing	2023
KOSGEB Entrepreneurship Training	2023
Circuit Design with Proteus	2025
Introduction to Basic Electronics and Robotics	2025
Robotic Coding with mBlock	2025
Image Processing Onramp	2025
MATLAB Onramp	2025
Computer Vision Onramp	2025
Arduino Programming with Projects	2025
Motors and Motor Control Circuits	2026
Fundamentals of Flight Mechanics	2026